Tamir Eliav

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Date of birth: 23.10.1987, Citizenship: Israeli, Family status: Married, 2 children

Education & Academic Positions

2022-present	**Post-Doctoral Researcher**, Max Planck Institute for Biological Intelligence, Seewiesen, Germany Vallentin lab, Neural Circuits for Vocal Communication Parental leave 04-10/2023
2015-2022	**PhD in Neuroscience**, Weizmann Institute of Science, Rehovot, Israel Advisor: Prof. Nachum Ulanovsky. GPA: 95.2/100 (class GPA: 91.5) Dissertation title: "Nonoscillatory phase coding and multiscale representation of very large environments in the bat hippocampus"
2013-2015	**Msc in Neuroscience**, Weizmann Institute of Science, Rehovot, Israel Advisor: Prof. Nachum Ulanovsky. Thesis grade: 98/100
2003-2006	**Bsc in Pure Mathematics**, Bar-Ilan University, Ramat-Gan, Israel Dean's Honors list, Faculty of Exact Sciences

ADDITIONAL TRAINING

2019	**Methods is Computational Neuroscience course** Marine Biology Laboratory, USA
2016	**The hippocampus: from circuits to cognition (Cajal Brain Prize course)** Bordeaux, France
2010-2011	**Graduate courses in Pure Mathematics** Bar-Ilan University, Israel

Teaching Experience

2017-2022	**Introduction to Matlab and data analysis** Teaching Assistant and guest lecturer. Feinberg Graduate School, Weizmann Institute of Science, Rehovot, Israel
2014-2019	**Bats, Brain and Memory** Lecturer Davidson Institute of Science Education, Rehovot, Israel

Publications

Maimon, S.R.*, **Eliav, T.***, Aljadeff, J., Shalev, A., Gronich, Y., Finger, N.M, Eveland, K.E., Moss, C.F., Las, L., Ulanovsky, N., 2026. Sparse-to-dense coding transformation between hippocampal subregions CA3 and CA1. ***Nature***

Palgi, S., Ray, S., Maimon, S.R., Waserman, Y., Ben-Ari, L., **Eliav, T.**, Tuval, A., Cohen, C., Keyyu, J., Ali, A.I., Mouritsen, H., Las, L., Ulanovsky, N., 2025. Head-direction cells as a neural compass in bats navigating outdoors on a remote oceanic island. ***Science***

Eliav, T.*, Maimon, S.R.*, Sarel A., Palgi S., Blum D., Las, L., Ulanovsky, N., 2025. Fragmented replay of very large environments in the hippocampus of bats. ***Cell***

Eliav, T.*, Maimon, S.R.*, Aljadeff, J., Tsodyks, M., Ginosar, G., Las, L., Ulanovsky, N., 2021. Multiscale representation of very large environments in the hippocampus of flying bats. ***Science***, 372, eabg4020
See also news & views:
Wood, E.R., Dudchenko, P.A., 2021. Navigating space in the mammalian brain. ***Science***, 372, 913–914

Eliav, T., Geva-Sagiv, M., Yartsev, M.M., Finkelstein, A., Rubin, A., Las, L., Ulanovsky, N. 2018. Nonoscillatory Phase Coding and Synchronization in the Bat Hippocampal Formation. ***Cell***, 175, 1119-1130.e15
See also news & views:
Trimper, J.B., Colgin, L.L., 2018. Spike Time Synchrony in the Absence of Continuous Oscillations. ***Neuron***, 100, 527–529
Bush, D., Burgess, N., 2019. Neural Oscillations: Phase Coding in the Absence of Rhythmicity. ***Current Biology***, 29, R55–R57

Awards, Fellowships & Grants

2025-2027	**Klaus Tschira Boost Fund**
2023	**Travel grant to attend SHRC meeting, Rosa Laura and Hartmut Wekerle Foundation**
2022	**Trainee Professional Development Award, Society for Neuroscience**
2022	**Travel grant for attending the Neuroethology meeting in Lisbon**
2022-2023	**Rothschild Postdoctoral Fellowship, awarded to the top 14 Israeli postdoctoral scholars in all scientific fields and engineering**
2022	**Postdoctoral Fellowship, Weizmann Institute of Science**
2022	**The John F. Kennedy Prize for outstanding PhD research, for recognition of academic excellence and scientific accomplishments , Weizmann Institute of Science**
2022	**FENS-Kavli network of excellence PhD Thesis Prize finalist**
2022	**FENS-IBRO/PERC Travel Grant for attending FENS forum in Paris**
2022	**Lee A. Segel Prize in Theoretical Biology**
2021	**FENS grant for attending SfN meeting**
2017-2021	**Maccabim Foundation Excellence Scholarship for Ph.D students**
2020	**Trainee Professional Development Award, Society for Neuroscience**
2020	**Otto Schwarz Academic Excellence Award**
2019-2021	**Horowitz foundation KKL–JNF Excellence Scholarship**
2019	**Israel psychobiology travel award for attending the SfN meeting in Chicago**
2019	**Boehringer Ingelheim Fonds travel grant for participating in the MCN course at MBL**
2019	**Caswell Grave Scholarship Fund travel grant for participating in the MCN course at MBL**
2018	**Heiligenberg Student Travel Award for attending the ICN meeting in Brisbane**
2017	**Best Poster Award, Israel Society For Neuroscience (ISFN) Annual Meeting, Eilat, Israel**
2014	**Best Poster Award, Israel Society For Neuroscience (ISFN) Annual Meeting, Eilat, Israel**
2005	**Dean's Honors List, Faculty of Exact Sciences, Bar-Ilan University**

Presentations

Invited Talks & Seminars

June 2023. *Multiscale representation of very large environments in the hippocampus of flying bats*. Invited talk, Bernstein Center for Computational Neuroscience retreat, Tutzing, Germany

May 2023. *Navigating out of the box*. Invited talk, Spring Hippocampal Research Conference, Verona, Italy

March 2023. *Nonoscillatory coding and multiscale representation of very large environments in the bat hippocampus*. Invited talk, Bernstein Center for Computational Neuroscience, Munich, Germany

Apr 2022. *Nonoscillatory coding and multiscale representation of very large environments in the bat hippocampus*. Neurobiology departmental seminar, Technion, Haifa, Israel

Jan 2022. *Multiscale representation of very large environments in the hippocampus of flying bats*. Israel Society for Neuroscience (ISFN) Annual Meeting, Eilat, Israel.

Dec 2021. *Nonoscillatory coding and multiscale representation of very large environments in the bat hippocampus*. Neurobiology departmental seminar, Tel-Aviv University, Tel-Aviv, Israel

Dec 2021. *Nonoscillatory coding and multiscale representation of very large environments in the bat hippocampus*. Biomedical Engineering departmental seminar, Ben-Gurion University, Beer-Sheva, Israel

Oct 2021. *Nonoscillatory coding and multiscale representation of very large environments in the bat hippocampus*. Brain Sciences departmental seminar, Weizmann Institute of Science, Rehovot, Israel

Oct 2021. *Nonoscillatory coding and multiscale representation of very large environments in the bat hippocampus*. Invited seminar, Friedrich Miescher Institute for Biomedical Research (FMI), Basel, Switzerland

Oct 2021. *Nonoscillatory coding and multiscale representation of very large environments in the bat hippocampus*. Invited seminar, Max Planck Institute for Ornithology, Seewiesen, Germany

Dec 2017. *Hippocampal representation of large-scale environments*. Swift Oral Presentation, Israel Society for Neuroscience (ISFN) Annual Meeting, Eilat, Israel.

Dec 2014. *Absence of low-frequency oscillations in the bat hippocampal formation*. Swift Oral Presentation, Israel Society for Neuroscience (ISFN) Annual Meeting, Eilat, Israel.

Jan 2014. *Absence of low frequency movement-related oscillations in the hippocampal formation of bats*. SPACEBRAIN meeting, Jaffa, Israel.

Posters

S., Maimon*, S.R., **Eliav, T.***, Las, L., Ulanovsky, N., Sparse versus dense coding of very large environments in hippocampal subregions CA3 and CA1. Society for Neuroscience (SfN) Annual Meeting, Chicago, USA (2024).

Palgi, S., Ray, S., Maimon, S.R., **Eliav, T.**, Tuval, A., Cohen, C., Las, L., Ulanovsky, N., Neurobiology of navigation in the real world: Head-direction cells serve as a neural compass in bats navigating outdoors on a remote oceanic island. Society for Neuroscience (SfN) Annual Meeting, Chicago, USA (2024).

Ray, S., Palgi, S., Maimon, S.R., **Eliav, T.**, Tuval, A., Cohen, C., Las, L., Ulanovsky, N., Neurobiology of navigation in the real world: Hippocampal spatial codes in bats navigating outdoors on a remote oceanic island. Society for Neuroscience (SfN) Annual Meeting, Chicago, USA (2024).

Palgi, S., Ray, S., Maimon, S.R., **Eliav, T.**, Tuval, A., Cohen, C., Las, L., Ulanovsky, N., Neurobiology of navigation in the real world: Head-direction cells serve as a neural compass in bats navigating outdoors on a remote oceanic island. iNAV: 5th Interdisciplinary Navigation Symposium, Italy (2024).

Eliav T, Maimon S.R, Sarel A, Palgi S, Blum D, Las L, Ulanovsky N, Fragmented replay of very large environments in the hippocampus of bats. Society for Neuroscience (SfN) Annual Meeting, San Diego, USA (2022).

Eliav T, Maimon S.R, Las L, Ulanovsky N, Fragmented replay of very large environments in the hippocampus of bats. International Congress of Neuroethology (ICN) Meeting, Lisbon, Portugal (2022).

Eliav T, Maimon S.R, Las L, Ulanovsky N, Neuronal replay sequences in the hippocampus of bats in a very large environment (200 meters). Federation of European Neuroscience Societies (FENS) Forum, Paris, France (2022).

Eliav T, Maimon S.R, Las L, Ulanovsky N, In search of neuronal replay sequences in the hippocampus of bats in a very large environment (200 meters). Society for Neuroscience (SfN) Annual Meeting, Chicago, USA (2021).

Maimon S.R*, **Eliav T***, Las L, Ulanovsky N, Sparse versus dense coding of very large environments in hippocampal subregions CA3 and CA1. Society for Neuroscience (SfN) Annual Meeting, Chicago, USA (2021).

Aljadeff, J, **Eliav, T**, Maimon, S.R, Ulanovsky, N, Tsodyks, M, An optimal spatial code in the hippocampus: environmental and behavioral effects. Computational and Systems Neuroscience (Cosyne) meeting (2021).

Aljadeff, J, **Eliav, T**, Maimon, S.R, Ulanovsky, N, Tsodyks, M, An optimal spatial code for large environments in the bat hippocampus. Bernstein Conference for Computational Neuroscience, Germany (2020).

Eliav T, Maimon S.R, Aljadeff J, Tsodyks M, Las L, Ulanovsky N, Representation of large-scale spaces in the hippocampus of flying bats. Israel Society For Neuroscience (ISFN) Annual Meeting, Eilat, Israel (2020).

Eliav T, Maimon S.R, Las L, Ulanovsky N, Representation of large-scale spaces in the hippocampus of flying bats, Society for Neuroscience (SfN) Annual Meeting, Chicago, USA (2019).

Eliav T, Las L, Ulanovsky N, Representation of large-scale spaces in the hippocampus of flying bats. Society for Neuroscience (SfN) Annual Meeting, San Diego, USA (2018).

Eliav T, Las L, Ulanovsky N, Representation of large-scale spaces in the hippocampus of flying bats. International Congress of Neuroethology (ICN) Meeting, Brisbane, Australia (2018).

Eliav T, Las L, Ulanovsky N, Representation of large-scale spaces in the hippocampus of flying bats. Israel Society For Neuroscience (ISFN) Annual Meeting, Eilat, Israel (2017).

Eliav T, Las L, Ulanovsky N, Representations of large-scale spaces in the hippocampus of flying bats. Society for Neuroscience (SfN) Annual Meeting, Washington D.C., USA (2017).

Eliav T, Las L, Ulanovsky N, Bat hippocampal recordings in large-scale environments. Israel Society For Neuroscience (ISFN) Annual Meeting, Eilat, Israel (2016).

Eliav T, Las L, Ulanovsky N, Towards bat hippocampal recordings in large-scale environments. Society for Neuroscience (SfN) Annual Meeting, San Diego, USA (2016).

Eliav T, Las L, Ulanovsky N, Towards bat hippocampal recordings in large-scale environments. Federation of European Neuroscience Societies (FENS) Forum, Copenhagen, Denmark (2016).

Eliav T, Las L, Ulanovsky N, Towards bat hippocampal recordings in large-scale environments, iNAV: 1st Interdisciplinary Navigation Symposium, Bad Gastein, Austria (2016).

Eliav T, Las L, Ulanovsky N, Towards bat hippocampal recordings in large-scale environments, International Congress of Neuroethology (ICN) Meeting, Montevideo, Uruguay (2016).

Eliav T, Geva-Sagiv M, Yartsev M, Finkelstein A, Rubin A, Las L, Ulanovsky N, Synchronicity without rhythmicity in the hippocampal formation of behaving bats, Israel Society For Neuroscience (ISFN) Annual Meeting, Eilat, Israel (2015).

Eliav T, Geva-Sagiv M, Yartsev M, Finkelstein A, Rubin A, Las L, Ulanovsky N, Synchronicity without rhythmicity in the hippocampal formation of behaving bats, Society for Neuroscience (SfN) Annual Meeting, Chicago, USA (2015).

Eliav T, Geva-Sagiv M, Yartsev M, Finkelstein A, Rubin A, Las L, Ulanovsky N, No movement-related oscillations in the entorhinal-hippocampal system of behaving bats despite low-frequency cellular resonance, Israel Society For Neuroscience (ISFN) Annual Meeting, Eilat, Israel (2014).

Eliav T, Geva-Sagiv M, Yartsev M, Finkelstein A, Rubin A, Las L, Ulanovsky N, No movement-related oscillations in the entorhinal-hippocampal system of behaving bats despite low-frequency cellular resonance, Society for Neuroscience (SfN) Annual Meeting, Washington D.C., USA (2014).

Eliav T, Geva-Sagiv M, Yartsev M, Finkelstein A, Rubin A, Las L, Ulanovsky N, No movement-related oscillations in the entorhinal-hippocampal system of behaving bats despite low-frequency cellular resonance, International Congress of Neuroethology (ICN) Meeting, Sapporo, Japan (2014).

Las L, **Eliav T**, Saraf-Sinik I, Vecht J, Ulanovsky N, Developing methods for multi-channel neural recording and stimulation in freely flying bats, International Congress of Neuroethology (ICN) Meeting, Sapporo, Japan (2014).

Eliav T, Geva-Sagiv M, Yartsev M, Finkelstein A, Rubin A, Las L, Ulanovsky N, No movement-related oscillations in the entorhinal-hippocampal system of behaving bats despite low-frequency cellular resonance, Federation of European Neuroscience Societies (FENS) Forum, Milan, Italy (2014).

Eliav T, Geva-Sagiv M, Yartsev M, Finkelstein A, Rubin A, Las L, Ulanovsky N, In search for oscillations and sonar phase-precession in the hippocampal formation of bats, Israel Society For Neuroscience (ISFN) Meeting, Eilat, Israel (2013).

Las L, **Eliav T**, Saraf-Sinik I, Vecht J, Ulanovsky N, Developing methods for multi-channel neural recording and stimulation in freely flying bats, Israel Society For Neuroscience (ISFN) Annual Meeting, Eilat, Israel (2013).

Las L, **Eliav T**, Saraf-Sinik I, Vecht J, Ulanovsky N, Developing methods for multi-channel neural recording and stimulation in freely flying bats, Society for Neuroscience (SfN) Annual Meeting, San Diego, USA (2013).

Outreach & Professional Development

Service and Outreach

2017-present	**Faculty of 1000 (F1000Prime)** Associate Faculty Member Writing recommendations for published scientific articles
2015-2022	**Student Committee for Inviting Outstanding Neuroscience researchers** Member Neurobiology Dept., Weizmann Institute of Science
2005-2006	**PERACH: mentoring program for social impact** Mentor Assisting 10-year-old child with ADHD educationally and socially

Professional Memberships

Society for Neuroscience (SfN), International Society for Neuroethology (ISN), Israel Society for Neuroscience (ISFN), Federation of European Neuroscience Societies (FENS)

Mentoring

2019	**Yohai Nirenberg** MSc rotation project in electrophysiology & data analysis
2018	**Daniel Rimon** MSc rotation project in electrophysiology & data analysis
2017	**Pritish Patil** MSc rotation project in electrophysiology & data analysis
2016	**Dan Blum** MSc rotation project in tracking animal behavior

Previous employment

2006-2011 **Israel Defense Forces** Embedded Software and algorithm R&D